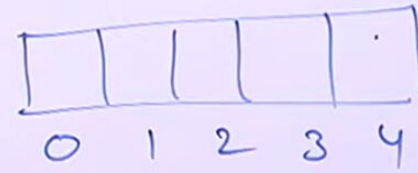
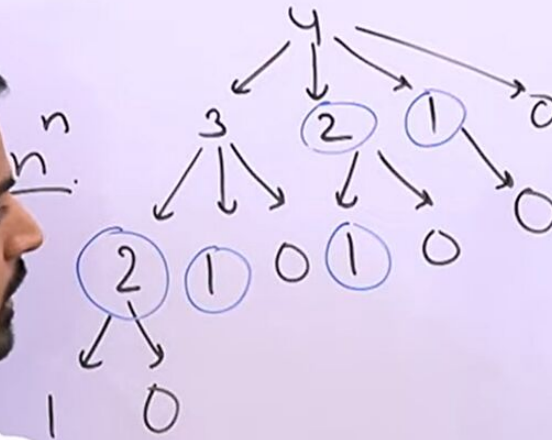


Cutting a rod (Dynamic Programming)

length :	1	2	3	4	5	6	7	8
price :	1	5	6	9	11	12	14	16



Cutting a rod (Dynamic Programming)

length:	1	2	3	4	5	6	7	8
price:	1	5	6	9	11	12	14	16

$O(N^2)$

$O(N)$

→ Price[0] + value[6]
→ Price[1] + value[4]
→ Price[2] + value[3]
→ ⋮

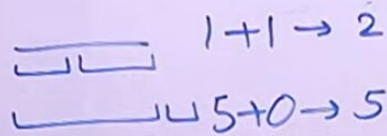
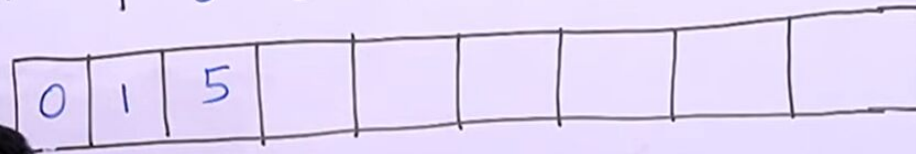
$$\text{value}(n) = \max_{i=0}^{n-1} (\text{price}[i] + \text{value}[n-i-1])$$

Price[5] + value[0]
↓
value[6]

which is value, this dimensional array named 'value'.

Cutting a rod (Dynamic Programming)

j → index :	0	1	2	3	4	5	6	7
i → length :	1	2	3	4	5	6	7	8
price :	1	5	6	9	11	12	14	16



$$6 - 0 - 1 \rightarrow 5$$

$$i - j - 1$$

Price[0] + value[6]
 Price[1] + value[5]
 Price[2] + value[4]

⋮

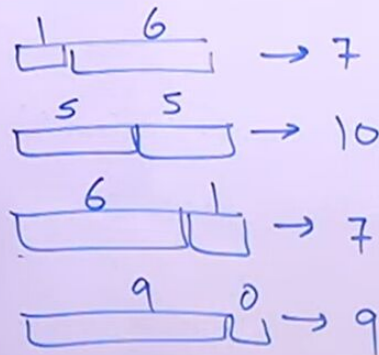
Price[5] + value[0]
 ↓
 value[6]

$$6 - 5 - 1 \rightarrow 0$$

Cutting a rod (Dynamic Programming)

j → index :	0	1	2	3	4	5	6	7
i → length :	1	2	3	4	5	6	7	8
price :	1	5	6	9	11	12	14	16

k →	0	1	5	6	10				
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$$6 - 0 - 1 \rightarrow 5$$

$$i - j - 1$$

Price[0] + value[6]
 Price[1] + value[5]
 Price[2] + value[4]

⋮

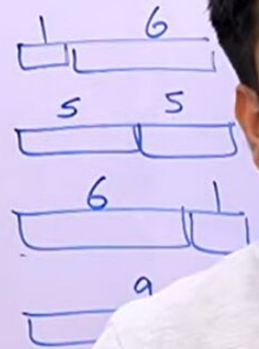
Price[5] + value[0]
 ↓
 value[6]

$$6 - 5 - 1 \rightarrow 0$$

Cutting a rod (Dynamic Programming)

$j \rightarrow$ index :	0	1	2	3	4	5	6	7
$i \rightarrow$ length :	1	2	3	4	5	6	7	8
price :	1	5	6	9	11	12	14	16

dp $\rightarrow$	0	1	5	6	10		15		20
------------------	---	---	---	---	----	--	----	--	----



$$6 - 0 - 1 \rightarrow 5$$

$$i - j - 1$$

$\text{Price}[0] + \text{value}[6]$
 $\text{Price}[1] + \text{value}[5]$
 $\text{Price}[2] + \text{value}[4]$
 $\vdots$

$\text{Price}[5] + \text{value}[0]$
 $\downarrow$
 $\text{value}[6]$

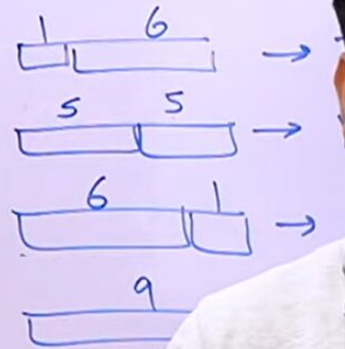
$6 - 5 - 1 \rightarrow 0$

The time complexity here is $O(N^2)$.

Cutting a rod (Dynamic Programming)

$j \rightarrow$ index :	0	1	2	3	4	5	6	7
$i \rightarrow$ length :	1	2	3	4	5	6	7	8
price :	1	5	6	9	11	12	14	16

dp $\rightarrow$	0	1	5	6	10	15		20
------------------	---	---	---	---	----	----	--	----



$$6 - 0 - 1 \rightarrow 5$$

$$i - j - 1$$

$$\text{Price}[0] + \text{value}[6]$$

$$\text{Price}[1] + \text{value}[5]$$

$$\text{Price}[2] + \text{value}[4]$$

⋮

$$\text{Price}[5] + \text{value}[0]$$

$$\downarrow$$

$$\text{value}[6]$$

$$6 - 5 - 1 \rightarrow 0$$

And the space complexity is $O(N)$.